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Omveer Singh

Data Analyst/Data Scientist

PROFILE SUMMARY

Quantitative researcher and PhD with 8+ years of international research experience, now transitioning into data analytics. Skilled in Python (Pandas), SQL, Tableau, ETL pipelines and A/B testing, with a track record in automated data validation, large-scale data analysis, and communicating complex findings to diverse audiences (20+ peer-reviewed publications). Seeking a Data Analyst role where I can deliver trusted, decision-ready insights.

TECH SKILLS

Data Analysis	Python (Pandas, NumPy), data cleaning & validation, statistical analysis, A/B testing
Databases & Querying	SQL, MySQL, REST APIs, web scraping
ETL & Cloud	ETL pipelines, Google Cloud
Visualisation & BI	Tableau, dashboards & business reporting, Matplotlib
Machine Learning	Scikit-learn, regression, classification
Engineering & Tools	Git/GitHub, C++, R, Linux, Docker, CI/CD, automated testing

DATA ANALYTICS TRAINING

Data Science Bootcamp **WBS Coding School** *Jul 2026 – Present*
Berlin, Germany

- Cleaned, reshaped and analysed real-world datasets with Python and Pandas, and presented the findings.
- Wrote SQL queries in MySQL to explore and combine data from several tables, and ran A/B tests to check whether changes had a real effect.
- Built interactive Tableau dashboards to answer business questions.
- Built an ETL pipeline: collected data through web scraping and APIs, cleaned it in Python and loaded it into a MySQL database on Google Cloud.
- Used Git and GitHub for version control in team projects.

Self-Directed Data Analytics Study *Oct 2025 – Jun 2026*
Darmstadt, Germany

- Worked through the Google Data Analytics Certificate, covering data cleaning, SQL and spreadsheets.
- Completed DeepLearning.AI's Supervised Machine Learning course (regression and classification).
- Built and published dashboards on Tableau Public.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher – Software & Data Systems **Goethe University Frankfurt** *Mar 2023 – Sep 2025*
Frankfurt am Main, Germany
Technologies: Python, C++, Data Validation, CI/CD, Git, Linux

- Built a QA framework in Python and C++ that automatically checks large simulation and detector datasets, so problems no longer had to be found by hand.
- Added these checks to the CI pipeline, so errors showed up as soon as the code changed.
- Compared detailed and simplified detector models and recommended which configuration to use, weighing accuracy against computing time.

Software Engineering Intern (GET_INvolved Programme) **GSI Helmholtz Centre / FAIR** *Jul 2022 – Dec 2022*
Darmstadt, Germany
Technologies: C++, CMake, Git, Linux

- Moved a large, old build system to modern CMake, which made the code easier to maintain.
- Wrote the developer documentation, which the collaboration adopted as its standard.

PhD Researcher – Data Analysis & Scientific Software **Aligarh Muslim University** *Dec 2015 – Dec 2022*
Aligarh, India

Technologies: Python, C++, Statistical Analysis, Machine Learning, Linux

- Thesis: analysed large simulated datasets to find out how well rare particles (low-mass vector mesons) could be measured, and used the results to recommend the best detector design.
- Wrote Python and C++ analysis code for large particle-physics datasets.
- Used statistical methods (Scaled Factorial Moments) to look for non-random patterns in collision data.
- Applied machine learning to pick out rare signals in noisy data.

SELECTED PROJECTS

Market-Entry Analysis (MySQL, Tableau)
Queried a 9-table marketplace database to decide whether an online tech retailer should partner with a Brazilian platform. Found that ∼93% of orders arrive on time and tech makes up ∼14% of revenue; recommended signing the deal and presented the results in a Tableau dashboard.

Playlist Clustering for a Music Startup (Python, Scikit-learn, Spotify API)
Built a K-Means model that grouped 5,000+ Spotify songs into 34 playlists of 50–250 songs each, to test whether machine learning can replace manual curation. Cleaned the data (duplicates, outliers, feature selection), compared 5 scaling methods and reduced 9 audio features with PCA while keeping 95% of the variance. Published the playlists via the Spotify Web API and explained the limits to a non-technical audience: the model makes good first drafts, but an expert should still review them.

Automated Cloud Data Pipeline for E-Scooter Demand Analysis (Python, MySQL, BeautifulSoup, REST APIs, Google Cloud)
Built an end-to-end ETL pipeline that scrapes city data from Wikipedia, pulls weather forecasts and flight schedules from REST APIs, and loads everything into a 5-table MySQL database on Google Cloud SQL. Deployed it as Cloud Run functions triggered by Cloud Scheduler, with duplicate-free re-runs and credentials stored in Secret Manager. Handled missing fields, empty responses, rate limits and timezone differences, and debugged production failures from Cloud Run logs.

Discount Strategy Analysis (Python, Pandas)
Cleaned three raw sales tables (malformed prices, 8,700+ duplicate rows) and tested whether discounts pay off. Sales stop rising above a 20% discount, and budget items get the deepest cuts but bring in only 5.9% of revenue; recommended capping discounts at 20% and shifting them to high-end products.

A/B Test of Homepage Buttons (Python, SciPy)
Compared four button designs across ∼100,000 homepage visits using a chi-square test with Bonferroni-corrected pairwise tests. White buttons clearly beat red ones (2.0–2.1% vs. 0.8–1.1% click-through rate); the two white versions showed no significant difference.

Smartwatch Activity Dashboard (BigQuery, Tableau)
Cleaned 960 days of personal Samsung Health data in BigQuery and built a Tableau dashboard; found the 10,000-step goal rate rose from under 30% to 73% in 2026.

QA Data Framework (Python, C++)
Framework that automatically checks large scientific datasets for errors, with quality checks built into the CI pipeline.

EDUCATION

Ph.D., Experimental High-Energy Physics — Aligarh Muslim University, India *Dec 2015 – Dec 2022*

M.Phil., M.Sc., B.Sc. (Hons.) in Physics — Aligarh Muslim University, India *Aug 2008 – Dec 2015*

CERTIFICATIONS

- Google Data Analytics Professional Certificate — Google (Coursera), in progress (6 courses completed)
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI (Coursera), 2025
- Databases and SQL for Data Science with Python — IBM (Coursera), 2025
- Python A–Z: Python for Data Science — Udemy, 2025
- R Programming A–Z: R for Data Science — Udemy, 2025
- View all certifications

ADDITIONAL INFORMATION

Languages: English C1, German A1
Work Authorisation: Eligible in Germany / EU

Publications: 20+ peer-reviewed papers (list)
Availability: Available immediately · Hybrid, on-site or remote



Darmstadt, October 22, 2026